Question Paper

Exam Date & Time: 19-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. (CORE STREAM) EXAMINATIONS - NOVEMBER 2024
SUBJECT: MAT 1171 - ENGINEERING MATHEMATICS -I

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) The following data gives the melting point of an alloy of lead and zinc, where t is the temperature in degree centigrade and p is the percentage of lead in alloy: (4)

p	40	50	60	70	80	90
t	184	204	226	250	276	304

Find the melting point of the alloy containing 84% of lead.

1B) Solve $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$. (3)

1C) Solve $y(2xy + e^x)dx - xe^x dy = 0$. (3)

2A) Use Runge-Kutta method of order four to find an approximate value of y at $x = 0.2, 0.4$, given $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$. Take $h = 0.2$, correct to four decimal places. (4)

2B) Solve $(D^2 - 2D + 1)y = xe^x sinx$. (3)

2C) Solve $(x+2)^2y'' - (x+2)y' + y = 3x + 4$. (3)

3A) Use Rayleigh power method to find the numerically largest eigenvalue and the corresponding eigenvector of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ by taking the initial eigenvector as $X^{(0)} = [1 \quad 0 \quad 0]^T$. Carry out four iterations, correct to two decimal places. (4)

3B) Use Newton's divided difference formula to find $y(6)$ from the following data: (3)

x	0	1	2	4	5
y	1	14	15	5	6

3C) Use Simpson's $\frac{1}{3}$ rule to find an approximate value for $\int_4^{5.2} \log x \, dx$ by taking six subintervals and correct to four decimal places. (3)

4A) Use Gram-Schmidt orthogonalization process to construct the orthonormal basis of $\mathbb{R}^3$ from the set of vectors $\{(1, -1, 0), (2, -1, -2), (1, -1, -2)\}$. (4)

4B) Use Taylor's series method to find an approximate value of y when $x = 4.1$ given $5x\frac{dy}{dx} + y^2 - 2 = 0$, $y(4) = 1$, up to 4th order terms. (3)

4C) Use Regula Falsi method to find the real root of the equation $2x - \log_{10} x = 7$ which lies in the interval $(3.5, 4)$. Carryout four iterations, correct to four decimal places. (3)

5A) Check whether the set of vectors $\{(3,0,2),(7,0,9)(4,1,2)\}$ forms a basis for $\mathbb{R}^3$. If so, express $(1,0,-1)$ as linear combination of elements in the basis. (4)

5B) Check the consistency and hence solve by Gauss elimination method (3)

$$4x_1 + x_2 + x_3 = 2$$

$$x_1 + 5x_2 + 2x_3 = -6$$

$$x_1 + 2x_2 + 3x_3 = -4$$

5C) Use Gauss-Seidel method to solve the system of equations (3)

$$6x + y + z = 20$$
$$x + 4y - z = 6$$
$$x - y + 5z = 7$$

with initial approximation as $x^{(0)} = y^{(0)} = z^{(0)} = 0$.

Carry out four iterations, correct to four decimal places.

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